Chunyu LIU

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Research Areas: Neurodegenerative disorders, Immune function, Multi-Omics Integration

EDUCATIONAL BACKGROUND

School of Public Health, Johns Hopkins University, Baltimore, MD 08/2023-expected 2027
PhD in the Department of Mental Health (Psychiatric Genomics); Concurrent MHS in the Department of Biostatistics

- Awards & Scholarship: 2024 The Morton Kramer Fund Award; 2025 Lucy Shum Memorial Scholar
- Relevant courses: Advanced data science, Advanced methods for statistical genetics and genomics, Statistical machine learning, etc.

School of Public Health, Peking University, Beijing, China 09/2021-06/2023
Master of Science in Epidemiology and Biostatistics

- Awards & Scholarship: 2022 Outstanding Research Award (5%)
- Relevant courses: Design and method of epidemiological studies, Genetic epidemiology, Methods in Biostatistics, Application of software packages, etc.

Peking University, Beijing, China 09/2016-06/2021
Bachelor of Medicine in Preventive Medicine; Double bachelor of Science in Psychological and Cognitive Sciences

- Awards & Scholarship: 2017 Excellent Social Practitioner at PKU (5%); 2017-2020 Excellent student scholarship (5%); 2021 Outstanding Research Award (5%)
- Relevant courses: Anatomy, Immunology, Pathology, Biochemistry, Nutrition and Food Hygiene, Functional Anatomy of CNS, Psychological Statistics, Developmental Psychology, Physiological Psychology, Cognitive Psychology, etc.

PUBLICATIONS AND PRESENTATIONS

* Denotes the first-author or equal contribution

Published

- Li C, **Liu C*** *et al*. Education, gender, and frequent body pain among middle-aged and older populations in the US, England, Mexico, China, and India. ***Pain*** 2024 https://doi.org/10.1097/j.pain.0000000000003349
- **Liu C*,** Lian Z, Li C, *et al*. How to Conduct Methodologically Rigorous Epidemiological Studies of the Chinese Famine of 1959-1961. ***J Epidemiol Community Health*** 2024 https://doi.org/10.1136/jech-2023-221875
- **Liu C***, Li C. C-reactive protein and cardiovascular diseases: a synthesis of studies based on different designs. ***Eur J Prev Cardiol*** 2023 https://doi.org/10.1093/eurjpc/zwad116
- **Liu C*,** Lv J *et al*. Tea consumption and cancer: a Mendelian randomization study (In Chinese). ***Chin J Epi*** 2023 https://doi.org/10.3760/cma.j.cn112338-20230217-00086
- Li C, Stebbins R, Nepport G, Carney C, **Liu C**, Ailello A, *et al*. Peripheral immune function and Alzheimer's disease: a living systematic review and meta-analysis. ***Mol Psychiatry*** 2023 https://doi.org/10.1038/s41380-023-02355-x
- **Liu C***, Li C. The need for appropriate 'age-balanced' controls and transparent reporting in Chinese famine studies: a re-analysis of the China Patient-centered Evaluative Assessment of Cardiac Events million persons project. ***Eur J Prev Cardiol*** 2022 https://doi.org/10.1093/eurjpc/zwac254
- Nie J, **Liu C**, Lv J *et al*. Independent and joint associations of tea consumption and smoking with Parkinson's disease risk in Chinese adults. ***J Parkinsons Dis*** 2022 https://doi.org/10.3233/JPD-223148
- **Liu C***, Meng X, Tang K *et al*. Early-life famine exposure and rheumatoid arthritis in Chinese adult populations. ***BMJ Open*** 2021 https://doi.org/10.1136/bmjopen-2020-043416
- **Liu C***, Cao M, Tang K *et al*. Association between sexual orientation and self-rated health: moderated mediation analysis of interpersonal relationship and household socioeconomic status. ***LGBT Health*** 2021 https://doi.org/10.1089/lgbt.2020.0320
- Zhu Y, **Liu C**, *et al*. Adult self-rated health and its influencing factors during COVID-19 epidemic (In Chinese). ***Health Educ Health Promot*** 2021 https://doi.org/10.16117/j.cnki.31-1974/r.202105441
- Chen H, **Liu C**, *et al*. Long-term care insurance beneficiaries' satisfaction and influencing factors—An analysis based on the survey data from Shangrao City, Jiangxi Province (In Chinese). ***Soc Secur Stud*** 2020 https://doi.org/10.3969/j.issn.1674-2982.2020.12.003

Conference papers and posters

- **Poster presentation in 2025 GSA,** Boston, "Links Between Burden of Common Infections and Cognitive Decline over More Than a Decade"

- **Poster presentation in 2025 GSA,** Boston, "Associations between objective sleep measures and proteomic aging: The ARIC Study"
- **Oral presentation in 2025 SLEEP,** Seattle, "Circadian Rest-Activity Rhythms and Epigenetic Age Acceleration in Older Adults"
- **Poster presentation in 2022 SER,** Chicago, "The use of Cohort Size Shrinkage Index (CSSI) to quantify regional famine intensity during the Chinese famine of 1959-61"
- **Poster presentation in 2020 Annual Meeting of Chinese Geriatrics Association,** Beijing, "Long-term care insurance beneficiaries' satisfaction and influencing factors—An analysis based on the survey data from Shangrao city, Jiangxi Province"

RESEARCH EXPERIENCE

Sleep and biological aging

Circadian rest-activity rhythms and epigenetic aging **10/2024-10/2025**

Lead researcher; Advisor: Dr. Brion Maher & Adam Spira, Johns Hopkins University

- Processed large-scale actigraphy time-series data to extract circadian rhythm features (RA, M10/L5, IS, IV).
- Investigating the association between circadian rest-activity rhythms and DNA methylation aging acceleration, using data from the epidemiologic catchment area (ECA) study, including interaction effects of age, sex, and BMI
- Abstract accepted for oral presentation at the 2025 SLEEP Annual Meeting
- Manuscript under reviewed in Nature aging.

Sleep, Immune Aging Biomarkers, and Cognitive Decline **10/2024-10/2025**

Lead researcher; Advisor: Dr. Brion Maher & Adam Spira, Johns Hopkins University

- Integrated sleep metrics with immune aging biomarkers (innate, adaptive, inflammatory markers) and longitudinal cognitive scores to model biological pathways linking sleep disruption to cognitive decline.
- Built reproducible pipelines for feature engineering, biomarker preprocessing, and longitudinal outcome modeling.
- Applied multivariable regression and linear mixed-effects models to assess associations and identify immune signatures most predictive of cognitive trajectories.
- Manuscript in preparation.

Sleep-disordered breathing, sleeping stages and proteomic aging **10/2023-Present**

Lead researcher; Advisor: Dr. Adam Spira, School of Public Health, Johns Hopkins University

- Integrated two waves of sleep data (AHI, ODI, SaO_2 nadir, REM/NREM architecture) with high-dimensional SomaScan proteomics to model biological aging.
- Computed four proteomic aging clocks (ARIC, Tanaka, Sathyan, Lehallier) and derived global and organ-specific proteomic age acceleration indices.
- Built reproducible R/Python pipelines for data cleaning, feature engineering, longitudinal modeling, and enrichment analysis.
- Applied multivariable and linear mixed-effects models to quantify the impact of sleep-disordered breathing and sleep-stage disruption on proteomic aging.

Genomics

Habenula spatial transcriptomics **10/2024-Present**

Lead researcher; Advisor: Prof. Leonardo Collado Torres, Lieber Institute for Brain Development

- Built scalable pipelines for processing and integrating high-dimensional spatial transcriptomics Visium HD datasets.
- Applied unsupervised clustering (FICTURE) and spatial representation learning to identify functional microregions.
- Performed multi-step batch correction, spatial modeling, and feature selection (cleaningY, HVG detection).
- Used cross-modality alignment to annotate spatial domains via scRNA-seq integration.
- Conducted graph-based cell–cell communication modeling (LIANA+) to infer signaling networks.
- Produced reproducible analyses using R/Python; collaborated on algorithm development and biological interpretation.

Immune Functional Polygenic Risk Score **10/2024-Present**

Lead researcher; Advisor: Dr. Brion Maher & Nilanjan Chatterjee, Johns Hopkins University

- Developed an immune-functional polygenic risk score by integrating genome-wide summary statistics with immune pathway and cell-type–specific annotations.
- Applied PRSet/PRSice-2 and custom pipelines to construct functionally informed SNP sets and evaluate their contribution to immune-related traits.
- Implemented QC, LD clumping, allele harmonization, and effect-size modeling to generate robust immune PRS estimates.
- Assessed predictive performance and biological relevance through association testing with immune biomarkers and aging-related phenotypes.

Selected Prior Projects

Systematic review of the immune system and Alzheimer's disease-related dementia 06/2022-06/2023

Research assistant; Advisor: Prof. Grace Noppert, School of Public Health, University of Michigan

- Co-designed a living systematic review to synthesize human studies relating peripheral immunity to ADRD;

screened and extracted data from around 300 studies to build a dataset of existing eligible studies
- Analyzed the study characteristics, summarized findings of available peripheral immune biomarkers, and visualized the relationship between various peripheral immune biomarkers and ADRD across studies
- Customized quality assessment protocol based on NOS and ROB tools, conducted quality assessment, and formulated recommendations for future studies based on epidemiological principles.

Tea consumption and multiple chronic diseases: Mendelian Randomization Analysis 09/2021-06/2023
Lead researcher; Advisor: Prof. Jun Lv, School of Public Health, Peking University
- Explored the relationship between tea consumption and multiple chronic diseases, including cardiovascular diseases, type 2 diabetes and cancer, using the China Kadoorie Biobank (CKB) of 0.5 million people
- Conducted GWAS in CKB using BOLT-LMM and METAL software to find genetic instrumental variables (SNPs); repeated analysis in UK Biobank (UKB) and Biobank Japan (BBJ) for instrument validation
- Selected SNPs from BBJ, UKB, and CKB for MR analysis; performed one-sample MR using CKB individual data and the linearity test of the effect; performed Two-sample MR, using GWAS summary data from CKB and BBJ
- Primary analyses showed that tea consumption could decrease the risk of stroke, but no use in other diseases.

Quantify regional famine intensity during the Chinese famine of 1959-61 03/2020-12/2021
Lead researcher; Advisor: Prof. L.H. Lumey, School of Public Health, Columbia University
- Explored methods to measure famine intensity and established potential dose-response relationships between prenatal famine exposure and outcomes of interest
- Used multiple-wave census and mortality data to model cohort shrinkage and generate excess mortality as intensity measures, and synthesized them for use in future Chinese famine studies
- Applied new measures to multiple empirical studies (CDC tuberculosis data in Sichuan province and China Health and Retirement Longitudinal study) and compared with previous findings.

INTERNSHIP

Beijing Haidian District Centers for Disease Control Intern 07/2020-10/2020
- Rotated across different departments to learn structures and operations of the organization, including Infectious Diseases, Chronic Diseases, Food Monitoring, Health Education
- Learned and experimented with the early warning network for infectious diseases
- Applied epidemiological research skills and gained hands-on experience in public health practices, including epidemiological investigation, scene sampling, questionnaire survey, and data entry

Beijing Shijitan Hospital Intern 01/2018-01/2019
- Rotated across multiple clinical departments, including Infectious Disease, Gastroenterology, Surgery
- Learned and practiced basic clinical skills and participated in case discussions and medical lessons
- Assisted in inquiry, case collation and biomedical examinations, including electrocardiogram, artery blood gas analysis, skin suture

PROFESSIONAL SKILLS

Computer skills: Statistical programming, Geo-coding, Data visualization, Figure fine-tuning (Adobe Illustrator)
Software skills: R, Python, Stata
Analytical skills: Regression, Survival analysis, Mediation analysis, Mendelian randomization, GWAS
Language skills: Chinese (native), English